

Mathematical Formulation of CBFPdFactorCylinder

1 Introduction

This document presents the mathematical formulation of the `CBFPdFactorCylinder` class, which implements a Control Barrier Function (CBF) based safety constraint for cylindrical obstacles. The factor ensures that a robot maintains a safe distance from obstacles while considering its velocity.

2 Notation

- E_{12} : Projection matrix onto xy-plane
- $\mathbf{p}_{\text{obs}}$: Observed obstacle position (projected)
- $\mathbf{p}_{p0}$: Relative position vector between robot and obstacle
- d : Euclidean distance between robot and obstacle
- d_{safe} : Safe distance threshold
- α : Control gain parameter
- $\mathbf{v}_i$: Velocity vector
- $\mathbf{v}_{i,\text{proj}}$: Projected velocity vector (xy-plane)
- pose_i : Robot pose (position and orientation)
- $\frac{\partial \mathbf{t}}{\partial \text{pose}_i}$: Jacobian of translation w.r.t. pose

3 Mathematical Formulation

3.1 Projection Matrix

The projection matrix E_{12} projects 3D vectors onto the xy-plane:

$$E_{12} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

3.2 Observation Processing

The observed point is projected onto the xy-plane:

$$\mathbf{p}_{\text{obs}} = E_{12} \cdot \mathbf{p}_{\text{input}}$$

3.3 Position Calculation

The relative position vector between the robot and obstacle:

$$\mathbf{p}_{p0} = E_{12} \cdot \mathbf{t}(\text{pose}_i) - \mathbf{p}_{\text{obs}}$$

where $\mathbf{t}(\text{pose}_i)$ extracts the translation component from pose i .

3.4 Distance Calculation

The Euclidean distance with numerical stability ($\epsilon = 10^{-6}$):

$$d = \max(\|\mathbf{p}_{p0}\|, \epsilon)$$

3.5 Velocity Projection

The velocity vector is projected onto the xy-plane:

$$\mathbf{v}_{i,\text{proj}} = E_{12} \cdot \mathbf{v}_i$$

3.6 Control Barrier Function (CBF)

The barrier function $h(\mathbf{x})$ represents the safety constraint:

$$h(\mathbf{x}) = \frac{1}{d_{\text{safe}}} - \frac{1}{d}$$

3.7 Time Derivative of CBF

The time derivative incorporates velocity information:

$$\dot{h}(\mathbf{x}) = \alpha \cdot \frac{\partial h}{\partial \mathbf{x}} \cdot \mathbf{v} = \alpha \cdot \frac{\mathbf{p}_{p0}^\top \cdot \mathbf{v}_{i,\text{proj}}}{d^3}$$

3.8 Error Function

The complete error function combines the barrier function and its time derivative:

$$\text{error} = h(\mathbf{x}) + \dot{h}(\mathbf{x}) = \left(\frac{1}{d_{\text{safe}}} - \frac{1}{d} \right) + \alpha \cdot \frac{\mathbf{p}_{p0}^\top \cdot \mathbf{v}_{i,\text{proj}}}{d^3}$$

4 Jacobian Matrices

4.1 Jacobian with respect to pose (H_1)

$$H_1 = \frac{\partial \text{error}}{\partial \text{pose}_i} = \frac{\partial h}{\partial \text{pose}_i} + \frac{\partial \dot{h}}{\partial \text{pose}_i}$$

4.1.1 Component 1: Derivative of barrier function

$$\frac{\partial h}{\partial \text{pose}_i} = \frac{\mathbf{p}_{p0}^\top}{d^3} \cdot E_{12} \cdot \frac{\partial \mathbf{t}}{\partial \text{pose}_i}$$

4.1.2 Component 2: Derivative of time derivative

$$\begin{aligned} \frac{\partial \dot{h}}{\partial \text{pose}_i} &= \alpha \cdot \frac{\partial}{\partial \text{pose}_i} \left(\frac{\mathbf{p}_{p0}^\top \cdot \mathbf{v}_{i,\text{proj}}}{d^3} \right) \\ &= \alpha \cdot \left[\frac{\partial(\mathbf{p}_{p0}^\top)}{\partial \text{pose}_i} \cdot \frac{\mathbf{v}_{i,\text{proj}}}{d^3} + \mathbf{p}_{p0}^\top \cdot \mathbf{v}_{i,\text{proj}} \cdot \frac{\partial}{\partial \text{pose}_i} \left(\frac{1}{d^3} \right) \right] \\ &= \alpha \cdot \left[\frac{\mathbf{v}_{i,\text{proj}}^\top \cdot E_{12} \cdot \frac{\partial \mathbf{t}}{\partial \text{pose}_i}}{d^3} - \frac{3(\mathbf{p}_{p0}^\top \cdot \mathbf{v}_{i,\text{proj}})}{d^5} \cdot \mathbf{p}_{p0}^\top \cdot E_{12} \cdot \frac{\partial \mathbf{t}}{\partial \text{pose}_i} \right] \end{aligned}$$

4.1.3 Complete pose Jacobian

$$H_1 = \frac{\mathbf{p}_{p0}^\top}{d^3} \cdot E_{12} \cdot \frac{\partial \mathbf{t}}{\partial \text{pose}_i} + \alpha \cdot \left[\frac{\mathbf{v}_{i,\text{proj}}^\top \cdot E_{12} \cdot \frac{\partial \mathbf{t}}{\partial \text{pose}_i}}{d^3} - \frac{3(\mathbf{p}_{p0}^\top \cdot \mathbf{v}_{i,\text{proj}})}{d^5} \cdot \mathbf{p}_{p0}^\top \cdot E_{12} \cdot \frac{\partial \mathbf{t}}{\partial \text{pose}_i} \right]$$

4.2 Jacobian with respect to velocity (H_2)

$$H_2 = \frac{\partial \text{error}}{\partial \mathbf{v}_i} = \alpha \cdot \frac{\mathbf{p}_{p0}^\top}{d^3} \cdot E_{12}$$

5 Constraint Handling

The error is constrained to be non-positive:

$$\text{error} = \begin{cases} 0 & \text{if } h(\mathbf{x}) + \dot{h}(\mathbf{x}) > 0 \\ h(\mathbf{x}) + \dot{h}(\mathbf{x}) & \text{otherwise} \end{cases}$$

When the error is positive (constraint violated), both Jacobians are set to zero:

$$H_1 = \mathbf{0}_{1 \times 6}, \quad H_2 = \mathbf{0}_{1 \times 3}$$

6 Dimensionality

- $\text{error} \in \mathbb{R}^1$ (scalar)
- $H_1 \in \mathbb{R}^{1 \times 6}$ (pose Jacobian)
- $H_2 \in \mathbb{R}^{1 \times 3}$ (velocity Jacobian)
- $\frac{\partial \mathbf{t}}{\partial \mathbf{pose}_i} \in \mathbb{R}^{3 \times 6}$ (translation Jacobian)

7 Implementation Notes

- Numerical stability is ensured by clamping the distance d to a minimum value of 10^{-6}
- The projection matrix E_{12} effectively creates a cylindrical safety region (ignoring z-axis)
- The factor implements a soft constraint that becomes active only when the safety boundary is approached
- The α parameter controls the aggressiveness of the velocity-dependent term